

Existence of Cities

EC3335, Set 03

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Spring 2027

Contents

- (i) Historical data
- (ii) Technological shifts
- (iii) *Why do cities exist?*
- (iv) Backyard production model

Housekeeping

PS01 will be posted later today

- Due on **Wednesday, October 12th** on canvas by **11:59p**
- Should be straight forward
- Though after today you won't be able to start question 5
- Must be submitted as a pdf

Note:

- Use the space provided on the pdfs
- If you need more space, use scratch paper
- Points will be deducted for messy work

Historical data

Historical data

More than **4 billion** people living in urban area globally [†]

Urban populations are now greater than rural populations

- Since **2007**
- Only **10%** of humans lived in urban areas in **1800**
- Only **1%** of land is categorize as urban area

By **2050**, **66%** of global populations will live in urban areas

[†] Although the definition of an "urban area" is not exactly **well defined**

Number of people living in urban and rural areas, World

Number of people living in urban and rural areas, based on each country's national definition.

Our World
in Data

Table

Line

Bar

Change country or region



Data source: United Nations Department of Economic and Social Affairs, Population Division (2025) - [Learn more about this data](#)

Note: Because the estimates of city and metropolitan areas are based on national definitions of what constitutes a city or metropolitan area, cross-country comparisons should be made with caution.

OurWorldinData.org/urbanization | CC BY



Explore the data →

Number of people living in urban and rural areas, World

Number of people living in urban and rural areas, based on each country's national definition.

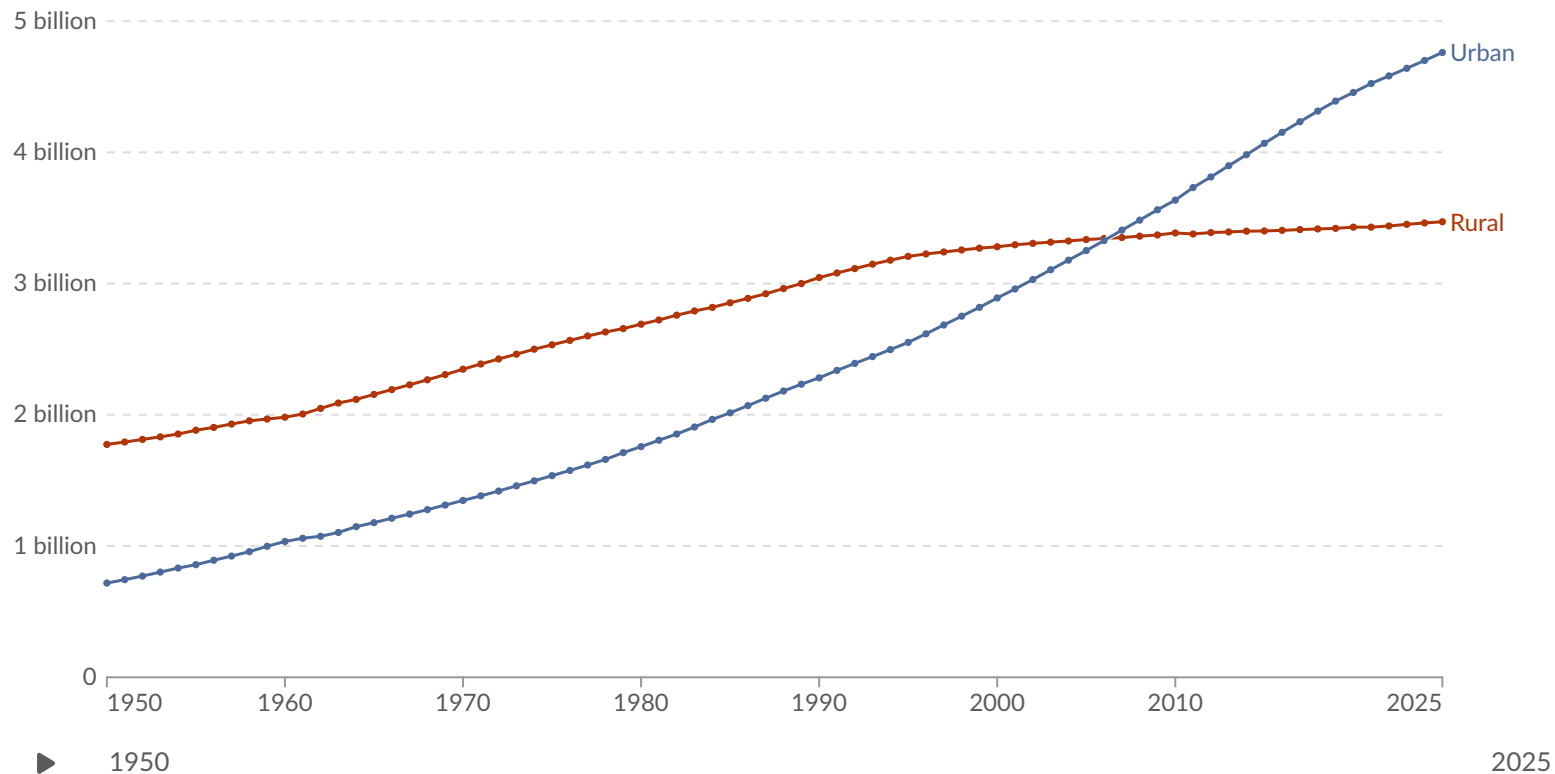
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Explore the data →

Share of population living in urban areas, 2025

Share of the population living in urban areas, based on each country's national definition.

Our World
in Data

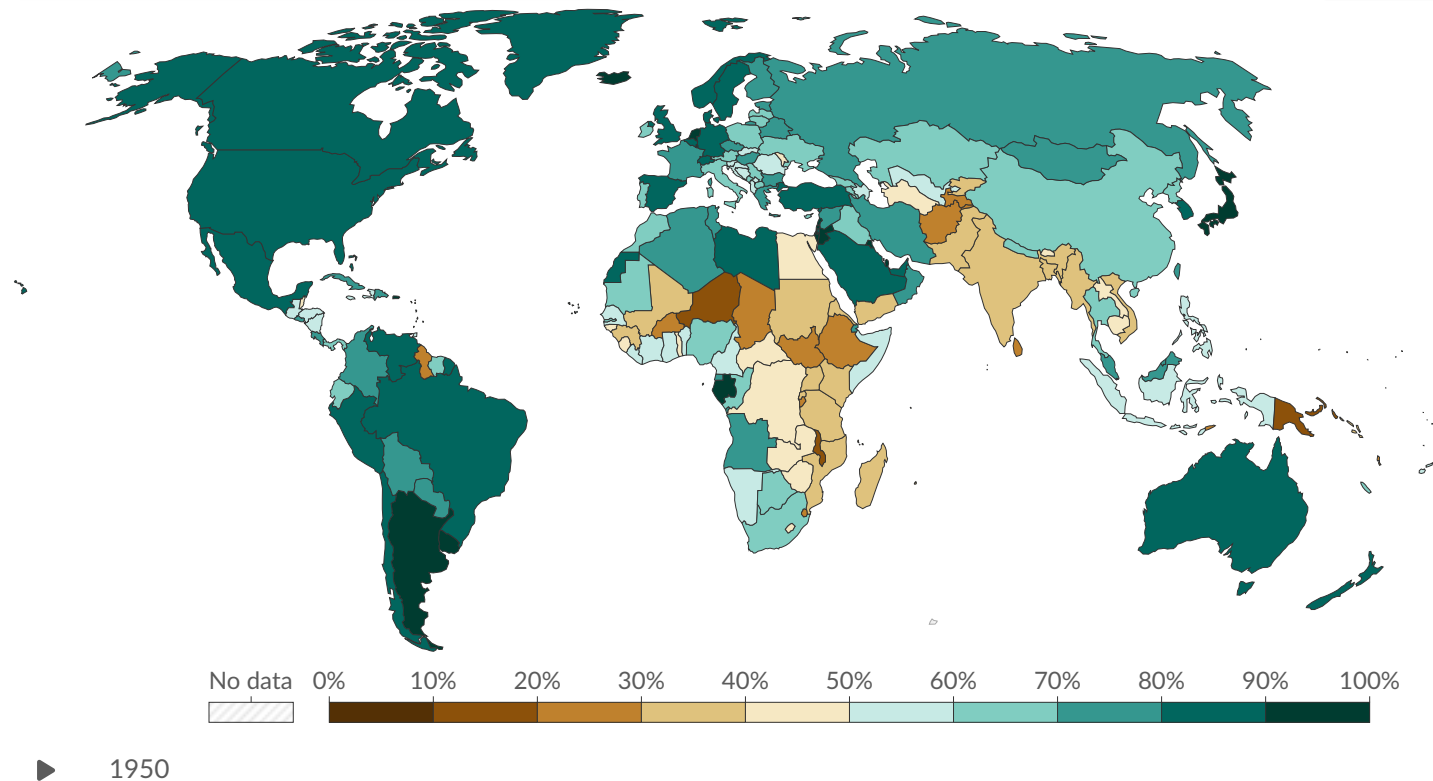
Table

Map

Line

Bar

Select countries



Data source: United Nations Department of Economic and Social Affairs, Population Division (2025) - [Learn more about this data](#)

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Explore the data →

Technological shifts

Technological shifts

Claim: In the past 200 years the world *has become* **flat**

- **1840s: Rail** becomes ubiquitous
 - Transportation costs 

Technological shifts

Claim: In the past 200 years the world *has become* **flat**

- 1840s: Rail becomes ubiquitous
 - Transportation costs ↓
- **1840s: Telegraph** is invented
 - Information costs ↓

Technological shifts

Claim: In the past 200 years the world *has become* **flat**

- 1840s: Rail becomes ubiquitous
 - Transportation costs ↓
- 1840s: Telegraph is invented
 - Information costs ↓
- **1870's: Telephone** is invented
 - Information costs ↓

Technological shifts

Claim: In the past 200 years the world *has become* **flat**

- 1840s: Rail becomes ubiquitous
 - Transportation costs ↓
- 1840s: Telegraph is invented
 - Information costs ↓
- 1870's: Telephone is invented
 - Information costs ↓
- **1908: Model T** production starts
 - Transportation costs ↓

Technological shifts

Claim: In the past 200 years the world *has become* **flat**

- 1840s: Rail becomes ubiquitous
 - Transportation costs ↓
- 1840s: Telegraph is invented
 - Information costs ↓
- 1870's: Telephone is invented
 - Information costs ↓
- 1908: Model T production starts
 - Transportation costs ↓
- **1914:** First commercial **airline flight**
 - Transportation costs ↓

Technological shifts

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- 1870's: Telephone is invented
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- 1908: Model T production starts
 - Transportation costs ↓
- 1914: First commercial airline flight
 - Transportation costs ↓
- **1980s:** Airline deregulation
 - Transportation costs ↓

Technological shifts

Claim: In the past 200 years the world *has become* **flat**

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- 1908: Model T production starts
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- 1914: First commercial airline flight
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- 1980s: Airline deregulation
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- **1990's:** The internet
 - Information costs ↓

Technological shifts

Claim: In the past 200 years the world *has become* **flat**

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- 1870's: Telephone is invented
 - Information costs ↓
- 1908: Model T production starts
 - Transportation costs ↓
- 1914: First commercial airline flight
 - Transportation costs ↓
- 1980s: Airline deregulation
 - Transportation costs ↓
- 1990's: The internet
 - Information costs ↓
- **2020s:** Post COVID
 - Transportation costs ↓
 - Information costs ↓
- Among many other technological changes

Why do cities exist?

Paradox: Why do cities exist?

Urban population trends and technological innovations seem **paradoxical**

The world is getting **flatter**, yet urban populations are **booming**

This **paradox** motivates the questions:

Why do cities exist in the first place?

Why have some cities succeeded?

Why have some cities failed?

We will draw insights from economic models to answer these questions

Paradox: Why do cities exist?

What do you think?

There is a pretty simple, one-word answer. **Any guesses?**

Trade.

Cities exist because individuals are not self-sufficient

Suppose there was **no trade**

- *What would we need for this to be true?*
- *Would this lead to no cities?*
- *Even with trade, when is it the case that households just trade amongst themselves?*

EC 201 Review 017: Trade

Definitions:

- **Absolute Advantage** (AA): An economic agent or entity has **AA** in exchange if they can produce more of the good in the same amount of time
- **Comparative Advantage** (CA) : An economic agent or entity has **CA** in exchange if they can produce the good at a lower *oppurtunity cost*
- **Production Possibilities Frontier** (PPF): All possible combinations of goods that an economic agent or entity can produce

EC 201 Review 017: Trade - example

Example: Suppose we have two countries, **A** & **B**, producing guns and oil

Each countries **PPF** is described by:

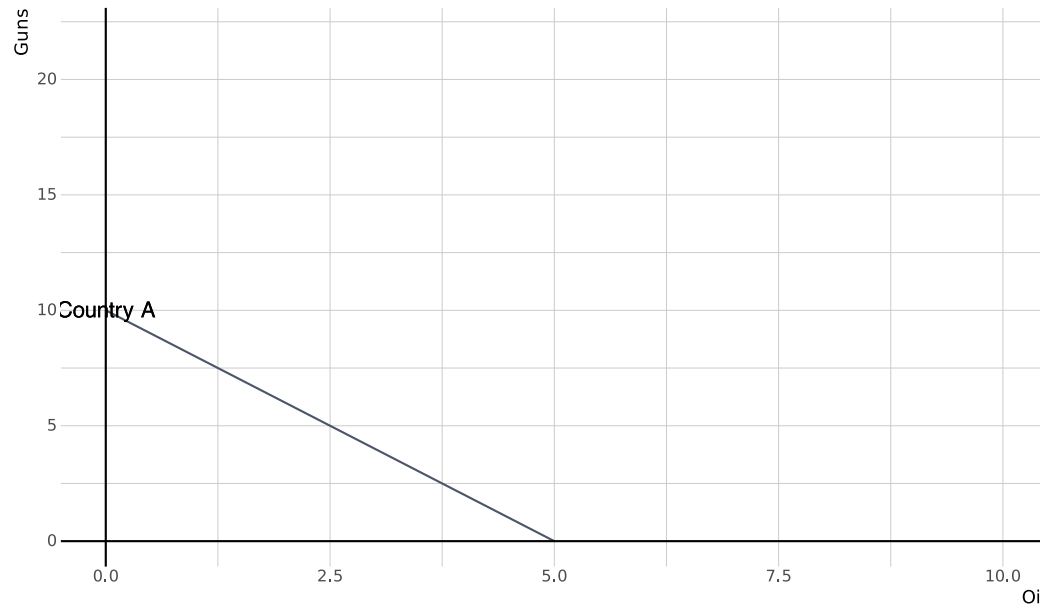
- **A:** $Guns_A = 10 - 2 * Oil_A$
- **B:** $Guns_B = 20 - 5 * Oil_B$

(i) Graph each countries **PPF**

(ii) Determine who has the AA and who has the CA in each good

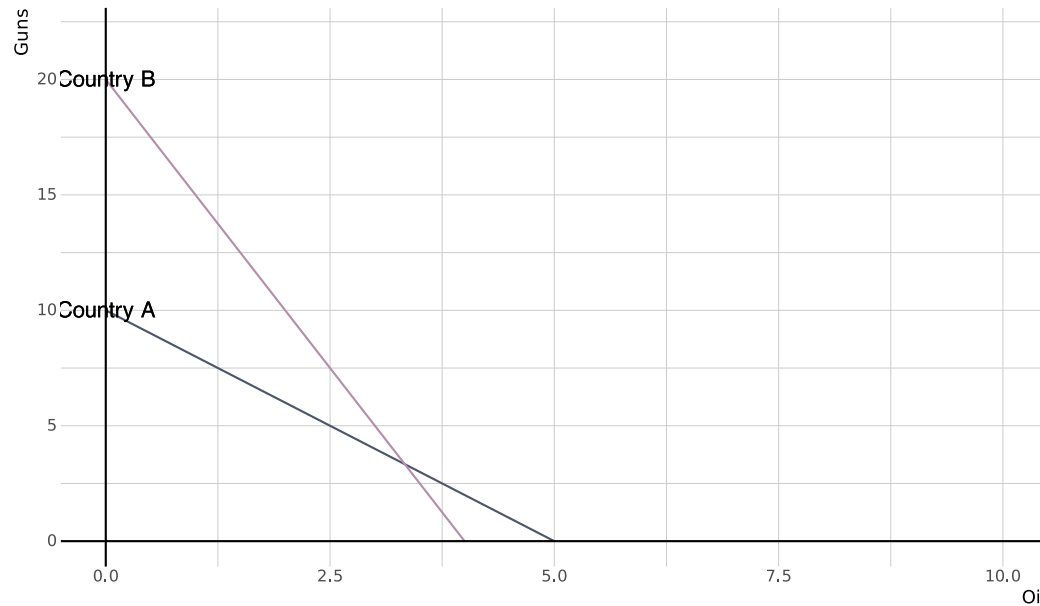
EC 201 Review 017: Trade - example

PPFs:



EC 201 Review 017: Trade - example

PPFs:



PPF Heuristics

When looking at PPFs, to determine:

(i) **AA**: Check intercepts

- Whoever has higher valued **intercept** has the **AA** in production

(ii) **CA**: Check slopes

- A **steeper slope** indicates CA on the vertical axis
- A **shallower slope** indicates CA on the horizontal axis

Why do cities exist?

We need land to produce food and resources; dense cities don't provide

Cities exist because of human **technology** has created systems of production and trade that defy the natural order

Three conditions must be satisfied:

(i). Agricultural surplus: People outside cities must produce enough food to feed themselves and city dwellers

(ii). Urban production: City dwellers must produce goods or services to exchange for food grown by rural workers

(iii). Transportation for exchange: Efficient transportation system to facilitate the exchange of food and urban products must exist

The technology in this case is what we call **trade**

Backyard production model

Backyard production model

A **simple** economic model to understand the economic incentives of cities

Consider a region that produces and consumers two products:

- **Bread**
- **Shirts**

Let's make three assumptions that would eliminate any incentives for households to geographically cluster

Relaxing each assumption will give us **intuition** about the formation of cities

Backyard production model

Model assumptions:

A01: There exist no differences in the productivity of land nor labor

- No comparative advantages

A02: Constant Returns to Scale (CRS) in **exchange** and **transportation**

- Per unit price to **trade** goods *is the same* no matter how much is traded
- No need for middle man firms to help with distribution

A03: CRS in **production**

- The per unit price of production is constant
- No economies of scale $\rightarrow$ households productivity = firm productivity

Backyard production model

These three assumptions would eliminate exchange and ensure each household is self sufficient

A01: Equal productivity

A02: CRS in exchange and transportation

A03: CRS in production

There is no incentive to specialization due to **transaction costs**[†]

Locational equilibrium would be **uniformly distributed** population

[†]transaction cost is the opportunity cost of the time required to exchange products

Backyard production model: Relax A01

Q: Is all land and labor equally productive?

A: Nope. Let's relax that assumption **Ex:**

- Soil may be more productive in certain regions; better climate
- One region specializes in bread while the other in shirts

Relaxing A01 will lead to differences in productivity across cities:

⇒ **comparative advantages**

⇒ specialization + trade

However, specialization and trade will not alone lead to urban growth

- Households are just as effective at trading as any firm

Relax A02

In absence of scale economies, households trade directly[†]

- **CRS in Exchange:** Implies households are just as efficient at executing trades as firms (no cost benefits to scaling)
- No reason to pay a firm to do so (and thus no reason to pay for density)

A trading firm will use productive inputs such as:

- Large truck for transportation
- Specialized workers

[†] Scale economies: *bigger* → *cheaper per unit*

Relax A02

To take full advantage of scale economies firms locate s.t. they minimize costs of distribution

ie. Build near river junctions, crossroads, ports

⇒ higher prices of land ⇒ density

Result: Trading cities

These represent the cities that **existed before the industrial revolution**

- Most city workers did not produce goods but distributed them
- Trade was risky; insurance, credit, banking and legal services sprouted

Relax A03

Now relax the final assumption **A03**

With economies of scale factories can outproduce households by lowering average costs

- Use indivisible inputs (machines)
- Allow workers to specialize

In order to pay for labor, a factory must pay such that they are indifferent between working in a factory and a rural area (**A1.**)

However, land scarcity **binds** and rents near the factory begin to increase

- Wages increase due to **locational equilibrium**

Factory area

Example:

Consider the shirt making factory:

- **Home** production: 20 p shirt
- **Factory**: 12 p shirt (economies of scale)

The factory locates in a town with 50 miles to east and west of villages

- 50 cents/mile to ship west.
- 20 cents/mile to ship east

Factory Towns

Market area: Area over which factories underprice home production

Under what condition will a consumer purchase the shirt from a factory?

$$\underbrace{p_f}_{\text{factory price}} + \underbrace{t \times d}_{\text{transit cost} = \text{cost p mile} \times \text{miles}} \leq \underbrace{p_h}_{\text{Home Price}}$$

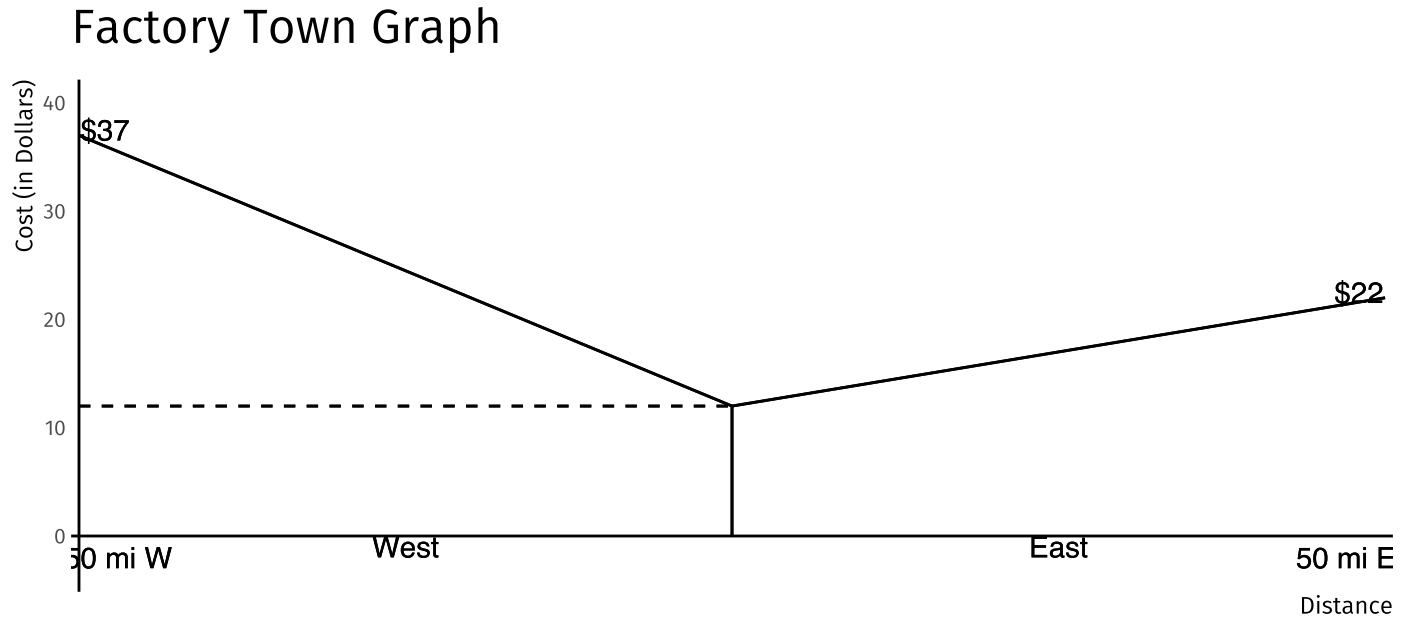
Questions:

1. Graph the cost of shirts throughout the entire region
2. Find the **market area** of the town
 - Find the sum of the **maximum distances** to the east and west that consumers will purchase the shirt from the factory

Regional Costs

Home production: 20 per shirt **Factory production:** 12 per shirt

Transportation costs: 50 cents/mile to ship west; 20 cents/mile east



Market Area Calculation

Market area depends on which side we are looking at. Let m denote miles

West: Consumers buy from factory if

$$12 + .5 * m_{west} \leq 20 \implies m_{west} \leq 16$$

East: Consumers buy from factory if

$$12 + .2 * m_{east} \leq 20 \implies m_{east} \leq 40$$

Market area:

$$40 + 16 = 56$$

Factory Towns

1. Would workers rather live **closer** or **further** from the factory?
 - **Closer!**
2. What happens to land-prices **close** to the factory?
 - They **increase**
3. What happens to **density**?
 - It will **increase**

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Data & History

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2. History
3. Paradox

hello

Existence

1. Why do Cities Exist?
2. Trade Basics
3. Factory Towns

Clustering

1. Zero Profit